SEC P vs NP log 2026-05-16

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 04:25 UTC

SEC P vs NP — daily log 2026-05-16

8 cycles recorded

Layer-Commutator Frobenius Discrepancy Separates Read-Twice BP for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Non-commutative operator discrepancy (Frobenius norm of layer-transition commutators, viewed as a Σ^b_1-witnessing measure in bounded arithmetic à la Cook–Krajíček) × Read-twice branching programs computing the inner-product-mod-2 function IP_2
- **Recorded**: 2026-05-16 20:13 UTC
- **Entry ID**: `0e06b5f9fc52`

Statement

Let P be a read-twice branching program with state set S ($|S|=s$) on n Boolean variables. For each of the $2n$ read-positions $r \in \{1,\dots,2n\}$ let $A_r \in \mathbb{R}^{s\times s}$ be the averaged transition matrix $A_r := (T_r^0 + T_r^1)/2$, where T_r^b is the 0/1 matrix sending state u to state v if reading bit b at position r drives $u\to v$. Define the layer-commutator Frobenius discrepancy $\rho(P) := \log_2\left(1 + \Sigma_{1\le r<q\le 2n} \|A_r A_q - A_q A_r\|_F^2 / s^2 \right)$. Conjecture: (a) $\rho(P) \le 6\cdot\log_2(s) + 4$ for every read-twice BP P; (b) for every read-twice BP P that computes IP_2 on $n \ge 4$ inputs, $\rho(P) \ge (1/8)\cdot n$. A single P violating either inequality refutes the conjecture; together they would force $\mathrm{size}(P) \ge 2^{n/48}$ for read-twice BPs computing IP_2.

Rationale

Bounded-arithmetic witnessing of BP membership relies on local invariants that are Σ^b_1-definable from the BP layer data; commutator energies of averaged layer matrices are exactly that kind of computable witness, while operator non-commutativity is the algebraic signature of the bilinear structure $IP_2(x,y)=\langle x,y\rangle_2$. Read-once BPs decompose into block-triangular layers where averaged transitions essentially commute on each reachable mass, but IP_2 forces every read-twice realization to mix x-reads and y-reads non-trivially. Because ρ is a Frobenius (L^2 spectral) quantity rather than a polynomial in the entries, the construction does not algebrize and the invariant is computed on BP structure (not the truth table), so it dodges Natural Proofs.

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1709.08890v2] Partial matching width and its application to lower bounds for branching programs - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [0710.5926v2]

Mod 2 cohomology of 2-local finite groups of low rank - [2603.27466v1] English translation of Frobenius' and Stickelberger's "On the theory of elliptic functions" - [0906.1662v2] Frobenius morphisms of noncommutative blowups

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12f9b311.py", line 211, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12f9b311.py", line 148, in run_trial
    brs_bp = build_brs_bp(n)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12f9b311.py", line 67, in build_brs_bp
    T0[u][v] = 1
    ~~~~~^^^
IndexError: list assignment index out of range
```

Judge reasoning

The test crashed with an IndexError in build_brs_bp before any trials produced data, so the pre-registered support condition cannot be evaluated. | next: Fix the indexing bug in build_brs_bp (state-set sizing for the BRS construction) and rerun the 150-trial sweep.

Matroid Matching of Term Co-Occurrence Graph Bounds k-CLIQUE Monotone DNF

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid matching theory (Lovász 1980 polymatroid matching theorem, applied to term co-occurrence graphs of monotone DNFs) × Monotone DNF size and Karchmer–Wigderson protocols for the k-CLIQUE indicator
- **Recorded**: 2026-05-16 20:46 UTC
- **Entry ID**: `02c57a92c88c`

Statement

For a monotone DNF $F = T_1 \vee \square \vee T_m$ on n variables, build the *term co-occurrence graph* $H_F = ([n], E_F)$ with $E_F = \{\{u,v\} : \exists\, i, \{u,v\} \subseteq T_i\}$, and define $\mu(F) = \mathrm{match}(H_F) / \lceil \log_2(1+m) \rceil$, where $\mathrm{match}(H_F)$ is the size of a maximum matching in H_F. We conjecture: (i) μ is subadditive under DNF conjunction-expansion, $\mu(F \wedge G) \le \mu(F)+\mu(G)$; (ii) $\mu(F) \le 4$ for every monotone DNF F of 'log-width' type, namely $m \le n^c$ and $\max_i |T_i| \le c \cdot \log_2 n$ for any fixed $c \ge 1$; (iii) for the canonical k-CLIQUE DNF $F_{v,k}$ on v vertices with $k = \lfloor \log_2 v \rfloor$ (variables = the $C(v,2)$ edges), $\mu(F_{v,k}) \ge v/(2 \cdot (\log_2 v)^2)$, i.e. unbounded as $v \to \infty$.

Rationale

Karchmer–Wigderson translates monotone formula size into communication on terms; matroid matching of the co-occurrence graph H_F is a representation-level polymatroid invariant capturing how 'expansive' the term overlap pattern is. Log-width poly-DNFs use few variables per term, keeping matchings tame, while every pair of vertices lies in some k-clique-term for k-CLIQUE, forcing H_F to be the complete graph K_v and matchings to scale linearly. Because μ reads the *syntactic* term structure rather than the truth table, it sidesteps Razborov–Rudich naturalness while staying squarely inside the KW monotone-protocol techniques.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [1503.06447v1] Sum-of-squares lower bounds for planted clique - [2405.01069v1] Oriented Ramsey numbers of graded digraphs - [2505.06446v1] Structured Prediction with Abstention via the Lovász Hinge - [1102.2932v2] Applications of Monotone Rank to Complexity Theory

Empirical Test

- exit code: `1`, elapsed: `0.36s`

```
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 9, 'conjecture_holds': False, '
CLIQUE bound failed: mu=0.0 < 0.4489652833957362 for v=6'}
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 9, 'conjecture_holds': False, '
CLIQUE bound failed: mu=0.0 < 0.4489652833957362 for v=6'}
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 9, 'conjecture_holds': False, '
CLIQUE bound failed: mu=0.0 < 0.4489652833957362 for v=6'}
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 9, 'conjecture_holds': False, '
CLIQUE bound failed: mu=0.0 < 0.4489652833957362 for v=6'}
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 9, 'conjecture_holds': False, '
CLIQUE bound failed: mu=0.0 < 0.4489652833957362 for v=6'}
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 9, 'conjecture_holds': False, '
CLIQUE bound failed: mu=0.0 < 0.4489652833957362 for v=6'}
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 9, 'conjecture_holds': False, '
CLIQUE bound failed: mu=0.0 < 0.4489652833957362 for v=6'}
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 9, 'conjecture_holds': False, '
CLIQUE bound failed: mu=0.0 < 0.4489652833957362 for v=6'}

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_eb505197.py", line 137, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_eb505197.py", line 137, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                              ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: The test reports conjecture_holds=False with an explicit counterexample: for the k-CLIQUE DNF at v=6, μ=0.0 < 0.449, violating clause (iii) of the pre | next: Recompute μ(F_{v,k}) by hand for small v to determine whether the lower bound v/(2·(log_2 v)^2) is fundamentally wrong or whether the matching/term-graph construction in the test mis-encodes the k-CLIQUE DNF.

Persistent H1 of Random Row Subclouds Bounds DISJ Communication

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Persistent homology / topological data analysis (Vietoris–Rips H_1 with total bar-length functional) × Randomized communication complexity of DISJOINTNESS (worst-case lower bound via sub-sampled row metric)
- **Recorded**: 2026-05-16 21:14 UTC
- **Entry ID**: `8fb2563d7c14`

Statement

Let M ∈ {0,1}^{N×N} be the communication matrix of a Boolean function (N = 2^n). For a uniformly random k-element subset S of row indices with k = ⌈√N⌉, view the rows {M[x,·] : x ∈ S} as a point cloud in {0,1}^N under Hamming distance, and let τ_PH(M;S) := Σ_{(b,d) ∈ Dgm_1(VR(S))} (d − b) denote the total persistence of the 1-dimensional bars of the Vietoris–Rips filtration of that sub-cloud, with all distances normalised by N. Conjecture: there is an absolute constant c > 0 such that for every Boolean matrix M, CC_R(M) ≥ c · log_2(1 + E_S[τ_PH(M;S)] · k); moreover E_S[τ_PH(M_DISJ_n;S)] = Ω(1) for every n ≥ 4, lifting the average-case topological signal to the worst-case bound CC_R(DISJ_n) = Ω(n).

Rationale

Persistent homology has been deployed in TDA but is essentially absent from communication-complexity lower bounds; unlike spectral or rank-based norms it is a multi-scale relational invariant (not a single-number truth-table statistic, so it dodges the natural-proofs trap) and does not algebrise because filtration birth/death values are not ring operations on a polynomial extension. The intuition is that high-CC matrices like DISJ scatter rows over many Hamming scales (producing long H_1 bars in subsamples) whereas low-CC matrices either cluster rows at few scales (e.g. Equality: all rows pairwise at Hamming distance 2, so every H_1 cycle is born and dies at the same filtration value, killing persistence) or admit small witnesses. Random-subset sampling matches the Impagliazzo–Wigderson–Hirahara avg-to-worst paradigm by deriving the worst-case CC_R bound from an expected topological invariant.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [2301.07191v3] A New Construction of the Vietoris-Rips Complex - [2605.00705v1] Cohomological properties of the Vietoris–Rips Complex of a Hypercube Graph - [2001.07588v8] Vietoris-Rips Persistent Homology, Injective Metric Spaces, and The Filling Radius - [0104048v1] A study of the link between cosmic rays and clouds with a cloud chamber at the CERN PS - [0104076v1] CLOUD: an atmospheric research facility at CERN

Empirical Test

- exit code: `0`, elapsed: `1.18s`


```
eed": 23, "n": 6, "family": "DISJ", "metric_name": "log2(1 + E_S[\u03c4_PH] * k)", "metric_value": 12.
TRIAL: {"seed": 23, "n": 6, "family": "IP", "metric_name": "log2(1 + E_S[\u03c4_PH] * k)", "metric_val
TRIAL: {"seed": 23, "n": 6, "family": "EQ", "metric_name": "log2(1 + E_S[\u03c4_PH] * k)", "metric_val
TRIAL: {"seed": 23, "n": 6, "family": "RAND", "metric_name": "log2(1 + E_S[\u03c4_PH] * k)", "metric_v
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TRIAL: {"seed": 37, "n": 4, "family": "EQ", "metric_name": "log2(1 + E_S[\u03c4_PH] * k)", "metric_val
TRIAL: {"seed": 37, "n": 4, "family": "RAND", "metric_name": "log2(1 + E_S[\u03c4_PH] * k)", "metric_v
TRIAL: {"seed": 37, "n": 5, "family": "DISJ", "metric_name": "log2(1 + E_S[\u03c4_PH] * k)", "metric_v
TRIAL: {"seed": 37, "n": 5, "family": "IP", "metric_name": "log2(1 + E_S[\u03c4_PH] * k)", "metric_val
TRIAL: {"seed":
```

Judge reasoning

Pre-registered support condition is not met: EQ mean reaches 13.30 at n=6, vastly exceeding the ceiling log2(6)+1 ≈ 3.58, and EQ matches IP exactly — clear evidence the metric is saturating at a k-determined ceiling rather than tracking communication complexity. The critic's challenge (small-n regime, trivially-satisfied O(log k) RHS, untested Ω(1) τ_PH claim) compounds this, so per the downgrade rule the apparent 100% support is not credible. | next: Re-run with a normalization that prevents me

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Non-Backtracking Spectral Gap Bounds Tseitin Resolution Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Non-backtracking (Hashimoto) operator spectrum on graphs × Resolution refutation length / KW-refutation game depth for Tseitin formulas
- **Recorded**: 2026-05-16 21:44 UTC
- **Entry ID**: `a58cbdcb6e05`

Statement

Let G be a connected d-regular graph (d≥3) on n vertices with odd-total Tseitin charge σ, let B(G) be the 2m×2m Hashimoto non-backtracking edge operator with spectral radius ρ(B), and define the graph invariant ν(G) := n · max(0, log((d−1)/ρ(B))). We conjecture (i) every Resolution refutation π of T(G,σ) satisfies |π| ≥ 2^(c·ν(G)) for an absolute constant c ≥ 1/16, and equivalently (via Pudlák's KW-refutation game) the refutation game has communication cost ≥ c·ν(G); (ii) ν(G) = O(1) for every graph that is not a vertex expander (any G with a balanced cut of size o(n) has ρ(B) → d−1, hence ν → 0). A single Tseitin instance refuted by Resolution in fewer than 2^(c·ν(G)) steps refutes the conjecture.

Rationale

The Hashimoto/Ihara non-backtracking spectrum captures geometric-group-theoretic cogrowth and Ramanujan-style expansion finer than the ordinary Laplacian, yet is computed by a single polynomial-time eigen-decomposition. KW-refutation games convert Resolution refutations into edge-charge-flow games on G whose communication cost is bounded by precisely the non-backtracking mixing rate of charges along geodesics, so ν(G) is the natural information-flow witness for the BSW expansion-vs-width principle. The invariant is structural (graph-only) rather than truth-table single-number, sidesteps Natural Proofs, and uses no polynomial extension of the Boolean ring, so it avoids Algebrization.

Novelty

- Judge: `NOVEL` over 3 arXiv hits

Top hits consulted: - [2209.05839v3] On bounded depth proofs for Tseitin formulas on the grid; revisited - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [2601.13117v2] The Case for Cardinality Lower Bounds

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b2cd758f.py", line 257, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b2cd758f.py", line 197, in run_trial
    B = build_non_backtracking_matrix(graph)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b2cd758f.py", line 129, in build_non_ba
    if graph[u][j] != v:
       ~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in build_non_backtracking_matrix before any data was produced, so neither the support nor the falsification condition can be evaluated. | next: Fix the indexing bug in build_non_backtracking_matrix (likely an off-by-one or assumption about graph[u]'s neighbor list length) and re-run the pre-registered protocol.

Matching Polynomial Mahler Measure Lower-Bounds Tseitin DPLL Tree Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matching polynomial root theory of Heilmann–Lieb–Godsil (algebraic graph theory: real-rootedness, Mahler measure of mu_G) × DPLL refutation tree size for Tseitin formulas on random cubic graphs
- **Recorded**: 2026-05-16 22:10 UTC
- **Entry ID**: `7351c55abee7`

Statement

For every connected 3-regular graph G on n vertices with odd-sum Z_2-charge c (so Tau(G,c) is unsatisfiable), let mu_G(x)=sum_k (-1)^k m_k(G) x^{n-2k} be the matching polynomial of G and let M(mu_G)=prod_{r: mu_G(r)=0, |r|>=1} |r| be its Mahler measure. We conjecture that for every such (G,c) the median DPLL refutation tree size T(Tau(G,c)) over 30 random VSIDS-free variable orderings satisfies log_2 T(Tau(G,c)) >= log_2 M(mu_G) - 1, equivalently T(Tau(G,c)) >= M(mu_G)/2 for every odd-charge cubic Tseitin instance. A single (G,c) with T < M(mu_G)/2 refutes the conjecture.

Rationale

Heilmann–Lieb constrains roots of mu_G to [-2√(d-1), 2√(d-1)] with multiplicities at the spectral edge controlled precisely by how 'tree-like' G fails to be, and Godsil's covering-tree identity makes M(mu_G) a quantitative measure of how many cycles obstruct a tree embedding. Since the Ben-Sasson–Wigderson width bound on Tseitin scales with edge expansion — which is exactly what forces matching-polynomial root mass near the spectral edge — the analytic complexity captured by Mahler measure should track DPLL depth. The matching polynomial has been deployed in statistical mechanics and zero-free regions of partition functions, but to our knowledge never as a Tseitin lower bound, giving a genuinely new bridge.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [0703195v3] Efficient Quantum Algorithm for Hidden Quadratic and Cubic Polynomial Function Graphs - [2209.05839v3] On bounded depth proofs for Tseitin formulas on the grid; revisited - [1806.00355v1] Mahler's work on Diophantine equations and subsequent developments - [0003039v1] Lieb-Thirring Inequalities

Empirical Test

- exit code: `-9`, elapsed: `78.75s`

```
(no output)
```

Judge reasoning

The test exited with returncode -9 (killed, likely OOM or timeout) and produced no RESULT line or data, so neither the support nor falsification condition can be evaluated. | next: Re-run with reduced n range (e.g. {8,10,12}) and fewer seeds, and add memory/time limits plus checkpointing so partial per-instance (T, M(mu_G)) data is persisted before any crash.

Mobius-Mertens Cancellation on Fourier Levels vs Total Influence

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Analytic number theory: the classical Mobius function mu(k) and Mertens-type cancellation applied to integer indices k=|S| labeling Fourier levels of a Boolean function × Total influence (average sensitivity) of Boolean functions on {-1,+1}^n, i.e. I(f) = sum_S |S|*f_hat(S)^2, the canonical Fourier-analytic complexity measure underlying KKL, Friedgut, and junta theorems
- **Recorded**: 2026-05-16 22:38 UTC
- **Entry ID**: `9ea129f53eed`

Statement

For every non-constant Boolean function f: {-1,+1}^n -> {-1,+1}, define the Mobius-Mertens level sum M(f) := sum_{k=1}^{n} mu(k) * W_k(f), where W_k(f) = sum_{|S|=k} f_hat(S)^2 is the level-k Fourier weight and mu is the number-theoretic Mobius function (mu(1)=1, mu(p)=-1 for primes, mu=0 on non-squarefree integers). Conjecture: |M(f)| * sqrt(1 + I(f)) <= 2 for an absolute constant 2, where I(f) = sum_S |S|*f_hat(S)^2 is total influence. A single instance with* |*M(f)*|sqrt(1+I(f)) > 2 falsifies it.

Rationale

The Mobius function exerts sqrt-type cancellation on its partial sums (Mertens phenomenon); applied to the discrete distribution {W_k(f)} on integer levels, it should produce cancellation tied to how spread-out the spectral mass is, which is exactly what total influence I(f) (the mean of the W_k-distribution) controls. A function concentrated at one level k can only avoid cancellation if mu(k) != 0, and the cost of high concentration is forced into I(f), suggesting a sqrt-law trade-off. To my knowledge no paper links classical mu-cancellation to Boolean influence bounds, giving a genuinely new analytic-number-theory x boolean-Fourier bridge.

Novelty

- Judge: `NOVEL` over 7 arXiv hits

Top hits consulted: - [2202.01071v5] Autocorrelation of the Mobius Function - [2206.12956v3] Result on the Mobius Function over Shifted Primes - [2003.09703v1] Variance function of boolean additive convolution - [0504289v3] Mertens' Proof of Mertens' Theorem - [2406.18700v4] Structure of sparse Boolean functions over Abelian groups, and its application to testing

Empirical Test

- exit code: `1`, elapsed: `0.12s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac9909ef.py", line 139, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac9909ef.py", line 114, in run_trial
    M, I, W_k = compute_metrics(f, n)
                ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac9909ef.py", line 89, in compute_metri
    fourier = fast_walsh_hadamard(f)
              ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac9909ef.py", line 38, in fast_walsh_ha
    even = fast_walsh_hadamard(f[:half] + f[half:])
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac9909ef.py", line 38, in fast_walsh_ha
    even = fast_walsh_hadamard(f[:half] + f[half:])
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac9909ef.py", line 38, in fast_walsh_ha
    even = fast_walsh_hadamard(f[:half] + f[half:])
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  [Previous line repeated 994 more times]
RecursionError: maximum recursion depth exceeded
```

Judge reasoning

The test crashed with a RecursionError in the Walsh-Hadamard transform implementation before producing any data, so neither the support nor falsification criterion can be evaluated. | next: Replace the recursive fast_walsh_hadamard with an iterative in-place implementation (or raise sys.setrecursionlimit and use numpy), then re-run; also explicitly include PARITY_n as a sanity check since |M(PARITY_5)|*sqrt(1+5) = sqrt(6) ≈ 2.449 would immediately falsify.

Column-Subset Discrepancy Lower-Bounds Disjointness Protocol Tree Leaves

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial discrepancy theory (Chazelle-Matousek-style hereditary discrepancy of column subsystems, with Beck-Fiala / Banaszczyk-type bounds) × Deterministic two-party communication complexity of DISJOINTNESS measured by leaf count of the canonical protocol tree on random sub-instances
- **Recorded**: 2026-05-16 23:03 UTC
- **Entry ID**: `96f06d91d088`

Statement

For a random {0,1}-matrix A in {0,1}^{n x n} with i.i.d. Bernoulli(1/2) entries (viewed as the communication matrix M_f of a DISJOINTNESS sub-instance restricted to row/column index sets X,Y of size n), let herdisc_k(A) be the hereditary discrepancy max_{S subset [n], |S|=k} min_{chi in {-1,+1}^S} ||A_S chi||_infty, computed greedily by Beck-Fiala rounding on the k chosen columns. Let L(A) be the number of leaves in the canonical DPLL-style deterministic protocol that, on inputs (x,y) in {0,1}^n x {0,1}^n with M_f indexed by A's support, splits on the lowest-index coordinate where x_i = y_i = 1 (declaring NOT-DISJOINT) or otherwise descends until exhaustion (declaring DISJOINT). We conjecture: for every n >= 4 and every k in [2, floor(log2 n)+2], log2 L(A) >= c * k * herdisc_k(A) / sqrt(k log n), with absolute constant c >= 1/8, falsified by any single (A, k, seed) triple where the inequality is violated.

Rationale

Chazelle-Matousek discrepancy bounds the 'unavoidable irregularity' of any 0/1 incidence system, and Bourgain-Kashin-style column-subset dispersion is precisely what protocol-tree pruning must overcome: each leaf of the deterministic DPLL-style DISJOINTNESS protocol corresponds to a sign-balanced column restriction. Linking hereditary discrepancy of column subsystems to leaf-count is a dispersion-flavored angle that has not been pursued (the existing discrepancy/CC literature focuses on global disc(M_f), not k-restricted herdisc). If true, this yields a dimension-free leaf lower bound bypassing rank-based methods.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1505.03110v1] Near-optimal bounds on bounded-round quantum communication complexity of disjointness - [0111062v2] One-way communication complexity and the Neciporuk lower bound on formula size - [9906008v2] A Lower Bound on the Average-Case Complexity of Shellsort - [2508.01937v1] An Improved Bound for the Beck-Fiala Conjecture - [1306.6081v3] An improvement of the Beck-Fiala theorem

Empirical Test

- exit code: 1, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e7a3676a.py", line 93, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e7a3676a.py", line 58, in run_trial
    k_values = range(2, math.floor(math.log2(n)) + 3)
                                             ^
UnboundLocalError: cannot access local variable 'n' where it is not associated with a value
```

Judge reasoning

The test crashed with an UnboundLocalError before any trials produced data, so neither support nor falsification can be assessed. No RESULT line was emitted and the critic challenges on the basis of the crash. | next: Fix the scoping bug by defining n inside run_trial (or passing it as an argument) before computing k_values, then rerun the full 30-seed × 10-n grid.

Zaslavsky Chamber Count Lower-Bounds AC^0 PARITY Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Hyperplane arrangement combinatorics (Zaslavsky chamber counting and Orlik-Solomon characteristic polynomial on the Boolean cube) × AC^0 circuit size and depth for PARITY
- **Recorded**: 2026-05-16 23:34 UTC
- **Entry ID**: `60678ef5bd8d`

Statement

For an AC^0 circuit C with gate set G on n inputs, define the gate-arrangement A(C) by associating to each gate g (fanin set S_g, type t_g ∈ {AND,OR}) the affine hyperplane H_g: Σ_{i ∈ S_g} ε_{g,i} x_i = θ_g, where θ_g = |S_g|−1/2 for AND and 1/2 for OR and ε_{g,i} encodes literal polarity. Define the discrete chamber count ch(C) := |{ (1[g(x)=1])_{g ∈ G} : x ∈ {0,1}^n }|, i.e. the number of distinct gate-output covectors realized on the Boolean cube (equal up to a factor 2 to the Zaslavsky restricted chamber count of A(C) on $\{0,1\}^n$). Then for every depth-d ≥ 2 AC^0 circuit C computing $PARITY_n$, $\log_2 ch(C) \geq (1/4) \cdot n^{1/(d-1)}$, and a single PARITY-computing circuit violating this bound at any (n,d) falsifies the conjecture.

Rationale

Nisan–Wigderson designs control the pairwise intersection structure of gate input sets to force PRG fooling; chamber count of the gate-arrangement is precisely the structural quantity counting how many such intersection patterns must be jointly realized inside any circuit. Because chamber count is a relational, gate-structural invariant — not a poly-time predicate on truth tables — it sidesteps Razborov–Rudich, while still inheriting Hastad-style sensitivity through the Boolean-cube restriction. Unlike single-number truth-table invariants (Mahler, Plünnecke, magnitude, etc.) which have all hit naturalness, ch(C) tracks the categorical incidence between gates and inputs, mirroring the $(S_i \cap S_j)$ condition central to NW designs.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2304.02770v2] Tight Correlation Bounds for Circuits Between AC0 and TC0 - [1705.10753v2] Tutte Polynomials of Symmetric Hyperplane Arrangements - [0011101v3] Morse theory, Milnor fibers and minimality of hyperplane arrangements - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2407.04826v1] Multi-strategy Based Quantum Cost Reduction of Quantum Boolean Circuits

Empirical Test

- exit code: `124`, elapsed: `240.14s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode=124) without producing any RESULT line or seed-level data, so the pre-registered support condition cannot be evaluated and the critic's CHALLENGE stands. | next: Reduce n range (e.g., n∈{6,8,10}) and/or seed count, and optimize chamber enumeration (cache covectors via bitsets) to bring runtime under the 240s budget before retrying.